. W_{Qq} q ,

$$W_{Qq} = -\frac{q^2 R}{4\pi\varepsilon_0 (l^2 - R^2)}. \quad (17)$$

,

$$W_q + W_Q + W_{Qq} + A = W_q. \quad (18)$$

(16)-(18)

$$W_Q = \frac{q^2 R}{8\pi\varepsilon_0 (l^2 - R^2)}. \quad (19)$$

2.6 **균일한 전기장 속의 도체 구** .

$r < R$,

$$q = \frac{4}{3}\pi r^3 \rho, \quad (20)$$

$$\Phi_E = 4\pi r^2 E. \quad (21)$$

$$\Phi_E = \frac{q}{\varepsilon_0}, \quad (22)$$

$$\vec{E} = \frac{\rho}{3\varepsilon_0}\vec{r}. \quad (23)$$

$\vec{r}$.

2.7 ,

. ,

$\overline{r_+}$ $\overline{r_-}$. → →

(23)

$$\overrightarrow{E_+} = \frac{\rho}{3\varepsilon_0}\overrightarrow{r_+}, \quad (24)$$

$$\overrightarrow{E_-} = -\frac{\rho}{3\varepsilon_0}\overrightarrow{r_-}. \quad (25)$$

$$\vec{E} = \frac{\rho}{3\varepsilon_0}(\overrightarrow{r_+} - \overrightarrow{r_-}) = \frac{\rho}{3\varepsilon_0}\vec{a}, \quad (26)$$

$\vec{a}$

.

2.8 .

(26) ,

$$\rho a = 3\varepsilon_0 E_0. \quad (27)$$

.

. δ

θ , a

$$\delta = a\cos\theta. \quad (28)$$

θ

$$\sigma = \frac{\rho V}{S} = \frac{\rho S \delta}{S} = \rho\delta. \quad (29)$$

(27)-(29)

$$\sigma = 3\varepsilon_0 E_0 \cos\theta. \quad (30)$$

2.9 , .

,